

Goal: To explore some fundamental characteristics of
g-expectations which are related to risk measures.

Contributions:

- (1) General Representation Lemma for generators of BSDEs under (A1) (A2) $\rightarrow$ Prop 2.3.
- (2) Under (A1) (A3), prove that if the static risk measure ρ^g , which is an operator, is convex (resp. coherent) risk measure, then the corresponding dynamic risk measure $(\rho_t^g)_{t \in [0, T]}$ which is an operator system, is a dynamic convex risk measure, and the generator g is independent of y and convex (sublinear) w.r.t. z .

Basis: BSDE (with unique adapted and square integrable solution, denoted $(Y_t(g, T, \xi), Z_t(g, T, \xi))_{t \in [0, T]}$)
(1.1) $Y_t = \xi + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s \cdot dB_s, 0 \leq t \leq T$.
 g - generator (Lipschitz in both y and z)
 $\xi, g(t, 0, 0)$ square integrable.
When g also satisfies $g(\cdot, y, 0) \equiv 0$ for any y , (A3)
 $Y_0(g, T, \xi)$, denoted by $Eg[\xi]$, is called g -
expectation of ξ ; $Y_t(g, T, \xi)$, denoted by $Eg[\xi | \mathcal{F}_t]$,
is called the conditional g -expectation of ξ .

Representation lemma for generators of BSDEs.

Prop 2.1 g - (A1) (A2). Let $1 \leq p \leq 2$.

For any triplet $(t, y, z) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d$.

$$g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} [Y_t(g, t+\varepsilon, y+z(B_{t+\varepsilon}-B_t)) - y] \quad (L^p)$$

$$\Leftrightarrow g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} E[\frac{1}{\varepsilon} \int_t^{t+\varepsilon} g(s, y, z) ds | \mathcal{F}_t] \quad (L^p)$$

Further studying shows that many problems on BSDEs are related to this kind of representation problem. In this section, we will establish a general Representation Lemma for generators of BSDEs under (A1) and (A2), which generalizes Proposition 2.3 of [3] and will be used frequently.

Lem 2.1 (Representation Lemma)

g - (A1) (A2). $1 \leq p < 2 \Rightarrow$ a.e. $t \in [0, T]$.

$$g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} [Y_t(g, t+\varepsilon, y+z(B_{t+\varepsilon}-B_t)) - y] \quad (L^p).$$

Prop 2.2 Let $q > 1, 1 \leq p < q, \mathcal{H}_{\mathcal{F}}^q(0, T; \mathbb{R}) := \{\psi \in \mathcal{M}_{\mathcal{F}}(\mathbb{R}); E \int_0^T |\psi_t|^q dt < \infty\}$. $\forall \psi \in \mathcal{H}_{\mathcal{F}}^q(0, T; \mathbb{R})$, we have

$$\psi_t = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds \text{ a.e. } t \in [0, T] \quad (L^p).$$

Proof. $\psi \in \mathcal{H}_{\mathcal{F}}^q(0, T; \mathbb{R})$. Fubini yields $\int_0^T E[|\psi_t|^q] dt < \infty$.
 $\Rightarrow E[|\psi_t|^q] < \infty$ a.e. $t \in [0, T]$. Lebesgue Lemma $\Rightarrow$.
 $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} E[|\psi_s|^q] ds = E[|\psi_t|^q]$ a.e. $t \in [0, T]$.

(18.4) Lemma [LEBESGUE]. Let f be a function in $\mathcal{L}_1([a, b])$. Then there is a set $E \subset (a, b)$ such that $\lambda(E' \cap [a, b]) = 0$ and

$$(i) \lim_{h \rightarrow 0} \int_x^{x+h} |f(t) - \alpha| dt = \lim_{h \rightarrow 0} \int_{x-h}^x |f(t) - \alpha| dt = |f(x) - \alpha|$$

for all $\alpha \in \mathbb{R}$ and all $x \in E$.

By Hölder, $\psi \in \mathcal{H}_{\mathcal{F}}^q(0, T; \mathbb{R}) \Rightarrow \int_0^T |\psi_t| dt \leq T^{1-\frac{1}{q}} \int_0^T |\psi_t|^q dt < \infty$ a.s.
 $\Rightarrow \int_0^T |\psi_t| dt \leq \int_0^T |\psi_t|^q dt < \infty$ a.s.

Again by 1.1, $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds = \psi_t$ a.e. a.s. (2.1)

Preliminaries.

1.2. Preliminaries. Let $T > 0$ be a fixed time horizon; let $(\Omega, \mathcal{F}, P)$ be a probability space and $(B_t)_{t \geq 0}$ be a d -dimensional standard Brownian motion on this space such that $B_0 = 0$; let $(\mathcal{F}_t)_{t \geq 0}$ be the augmented natural filtration generated by $(B_t)_{t \geq 0}$ and satisfy the usual conditions. Let $\mathcal{M}_{\mathcal{F}}(\mathbb{R}^n)$ denote the space of all $\mathbb{R}^n$ -valued, $(\mathcal{F}_t)$ -progressively measurable processes. We set $\mathcal{H}_{\mathcal{F}}^2(0, T; \mathbb{R}^n) := \{\psi \in \mathcal{M}_{\mathcal{F}}(\mathbb{R}^n); \|\psi\|_{\mathcal{H}}^2 := E[\int_0^T |\psi_t|^2 dt] < \infty\}$, $\mathcal{S}_{\mathcal{F}}^2(0, T; \mathbb{R}) := \{\psi \in \mathcal{M}_{\mathcal{F}}(\mathbb{R}); \psi \text{ is continuous and } E[\sup_{0 \leq t \leq T} |\psi_t|^2] < \infty\}$, $L^2(\mathcal{F}_T) := \{\xi; \xi \text{ is } \mathcal{F}_T\text{-measurable random variable, } E[\xi^2] < \infty\}$.

The generator g of a BSDE is a function $g: \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ such that $(g(t, y, z))_{t \in [0, T]}$ is progressively measurable for each $(y, z) \in \mathbb{R} \times \mathbb{R}^d$, and g also satisfies the following usual assumptions (A1) and (A2):

- (A1) There exists a constant $K \geq 0$ such that $dP \times dt$ -a.s., $\forall y_1, y_2, z_1, z_2$,
 $|g(t, y_1, z_1) - g(t, y_2, z_2)| \leq K(|y_1 - y_2| + |z_1 - z_2|)$.
- (A2) The process $(g(t, 0, 0))_{t \in [0, T]} \in \mathcal{H}_{\mathcal{F}}^2(0, T; \mathbb{R})$.
- (A3) $dP \times dt$ -a.s., for any $y \in \mathbb{R}, g(\cdot, y, 0) \equiv 0$.
- (A4) P -a.s., for any $y \in \mathbb{R}, z \in \mathbb{R}^d, t \mapsto g(t, y, z)$ is continuous.

Let (A1) and (A2) hold for g . By [16], for each $\xi \in L^2(\mathcal{F}_T)$, (1.1) has a unique solution in $\mathcal{S}_{\mathcal{F}}^2(0, T; \mathbb{R}) \times \mathcal{H}_{\mathcal{F}}^2(0, T; \mathbb{R}^d)$, which is denoted by

$$(Y_t(g, T, \xi), Z_t(g, T, \xi))_{t \in [0, T]}.$$

We recall the notions of g -expectation and conditional g -expectation and some properties given in Peng [17]. In the following Definitions 1.1 and 1.2 and Lemma 1.1, g is assumed to satisfy (A1) and (A3).

Def. 1.1. $Eg[\cdot]: L^2(\mathcal{F}_T) \mapsto \mathbb{R}$ is defined by

$$Eg[\xi] := Y_0(g, T, \xi).$$

Def. 1.2. $Eg[\cdot | \mathcal{F}_t]: L^2(\mathcal{F}_T) \mapsto \mathbb{R}$ is defined by

$$Eg[\xi | \mathcal{F}_t] := Y_t(g, T, \xi)$$

which is the unique r.v. η in $L^2(\mathcal{F}_t)$ s.t.

$$Eg[\xi | A] = Eg[\eta | A] \text{ for all } A \in \mathcal{F}_t.$$

Lem 1.1. (i) $\forall$ constant $c, Eg[c] = c$.

(ii) $X_1 \geq X_2$ a.s. $\Rightarrow Eg[X_1] \geq Eg[X_2]$

(iii) $X_1 \geq X_2$ a.s., $P(X_1 > X_2) > 0 \Rightarrow Eg[X_1] > Eg[X_2]$

(iv) g i.d. of $y \Rightarrow Eg[X + \eta | \mathcal{F}_t] = Eg[X | \mathcal{F}_t] + \eta$.
 $\forall X \in L^2(\mathcal{F}_T), \forall \eta \in L^2(\mathcal{F}_t)$.

Hence, $\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds = \psi_t$ a.s., a.e. $\nearrow$ 换顺序是为了固定 ω , 把 $\psi_s(\omega)$ 当成确定函数处理.
 $\exists S \in [0, T], \lambda([0, T] \setminus S) = 0$, and for each S
 $(2.2) \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds = \psi_t$ a.s.
 $(2.3) E[|\psi_t|^2] < \infty, \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} E[|\psi_s|^2] ds = E[|\psi_t|^2]$
 $\forall t \in S, \exists \delta_t > 0$ s.t.
 $(2.4) \frac{1}{\varepsilon} \int_t^{t+\varepsilon} E[|\psi_s|^2] ds \leq E[|\psi_t|^2] + 1, \forall \varepsilon \in (0, \delta_t)$
Set $X_t^\varepsilon := |\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds|$.
 $\forall N > 0, \text{Hölder} + \text{Fubini} + (2.4) \Rightarrow$

$$\begin{aligned} \int_{\{X_t^\varepsilon > N\}} |\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds|^p dP &\leq \int_{\{X_t^\varepsilon > N\}} \frac{1}{N^{p-1}} |\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds|^p dP \\ &= \int_{\{X_t^\varepsilon > N\}} \frac{1}{N^{p-1}} [\frac{1}{\varepsilon} \int_t^{t+\varepsilon} |\psi_s|^p ds] dP \\ &\leq \frac{1}{N^{p-1}} E[\frac{1}{\varepsilon} \int_t^{t+\varepsilon} |\psi_s|^p ds] \\ &\leq \frac{1}{N^{p-1}} [E[|\psi_t|^p] + 1]. \end{aligned}$$

Thus, $\{|\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds|^p; \varepsilon \in (0, \delta_t)\}$ uniformly integrable.
With (2.2). $\forall t \in S$.

$$\psi_t = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \psi_s ds \quad (L^p). \quad \square$$

Proof of Lem 2.1 (Lem 2.1 help us to find the pricing mechanism - g).

$$((g(t, 0, 0))_{t \in [0, T]}) \in \mathcal{H}_{\mathcal{F}}^2(0, T; \mathbb{R}), g\text{-(A1)}$$

$\Rightarrow \forall (y, z) \in \mathbb{R} \times \mathbb{R}^d, (g(t, y, z))_{t \in [0, T]} \in \mathcal{H}_{\mathcal{F}}^2(0, T; \mathbb{R})$.

So $\forall 1 \leq p < 2, (y, z) \in \mathbb{R} \times \mathbb{R}^d$. By Prop 2.2 + Jensen's Ineq.

We have,

$$(2.6) g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} E[\frac{1}{\varepsilon} \int_t^{t+\varepsilon} g(s, y, z) ds | \mathcal{F}_t] \text{ a.e. } t \in [0, T] \quad (L^p)$$

Thus, Lem 2.1 follows from (2.6) and Prop 2.1 $\square$.

Translation invariance, convexity, subadditivity and positive homogeneity.

Main difference:

- use $E_g[\cdot]$ instead of $E_g[\cdot | \mathcal{F}_t]$ to describe the character of the generator of g .
- Only under the usual assumption (A1) (A3)

New notation:

$$S_y^z(g) := \{t \in [0, T] \mid g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}_t(g(t, y + \varepsilon z, z) - g(t, y, z)) \mid \mathcal{F}_t\}.$$

If g i.d. of y , $\forall z \in \mathbb{R}^d$.

$$S^z(g) := \{t \in [0, T] \mid g(t, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E}_t(g(t, \varepsilon z, z) - g(t, 0, z)) \mid \mathcal{F}_t\}.$$

• Translation invariance for g -expectation.

Now we investigate the inverse problem of Lem 1.1 (iv).

Thm 3.1 g - (A1) (A3). The following 3 statements are equivalent:

- $E_g[\xi + c] = E_g[\xi] + c$, $\forall \xi \in L_T^2$, $c \in \mathbb{R}$. (Trans. Inv.)
- $\forall \xi \in L^2(\mathcal{F}_T)$, $c \in \mathbb{R}$, $\forall t \in [0, T]$,
 $E_g[\xi | \mathcal{F}_t] = E_g[\xi | \mathcal{F}_t] + c$, P -a.s.
- g i.d. of y .

Proof. (iii) $\Rightarrow$ (i): Lem 1.1 (iv); (ii) $\Rightarrow$ (i): trivial.

(i) $\Rightarrow$ (iii): $g^c(t, y, z) := g(t, y + c, z) \forall t \in [0, T], y \in \mathbb{R}, z \in \mathbb{R}^d$.

So g^c satisfies (A1) (A2) (A3) (A2) is automatically satisfied.

$\forall X \in L^2(\mathcal{F}_T)$. By the uniqueness of solution of BSDE,

$$(\Upsilon_t(g^c, T, X + c), Z_t(g^c, T, X + c))_{t \in [0, T]}$$

$$= (\Upsilon_t(g, T, X) + c, Z_t(g, T, X))_{t \in [0, T]}$$

$$\Rightarrow E_{g^c}[X + c] = E_g[X] + c = E_g[X + c]. \text{ (By (i))}$$

$$\Rightarrow \forall c \in \mathbb{R}, E_{g^c}[\xi] = E_g[\xi], \forall \xi \in L_T^2. \quad (3.1)$$

$$E_g[\cdot] \text{ is } L^2 \text{ continuous} \Rightarrow E_g[\xi + c] = \lim_{n \rightarrow \infty} E_g[(1 - \frac{1}{n})\xi + c] \\ = E_g[\lim_{n \rightarrow \infty} (1 - \frac{1}{n})\xi] + c = E_g[\xi] + c$$

$$\text{Hence, } E_g[\xi + c] \leq E_g[\xi] + c, \forall \xi \in L^2(\mathcal{F}_T), \forall c \in \mathbb{R}. \quad (3.9)$$

$$\text{Easily we obtain, } E_g[\xi] \leq E_g[\xi + c] - c, \forall \xi \in L^2(\mathcal{F}_T), \forall c \in \mathbb{R}. \quad (3.10)$$

$$\Rightarrow E_g[\xi + c] = E_g[\xi] + c, \forall \xi \in L^2(\mathcal{F}_T), \forall c \in \mathbb{R}. \quad (3.11)$$

Step 2: Prove that $E_g[\cdot | \mathcal{F}_t]$ convex. That is, $\forall \xi, \eta \in L^2(\mathcal{F}_T), \alpha \in [0, 1]$

$$\forall t \in [0, T], E_g[\alpha \xi + (1 - \alpha)\eta | \mathcal{F}_t] \leq \alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]. \quad (3.12)$$

$$\text{Set } A := \{E_g[\alpha \xi + (1 - \alpha)\eta | \mathcal{F}_t] > \alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]\} \in \mathcal{F}_t.$$

Suppose $P(A) > 0$:

$$1_A E_g[\alpha \xi + (1 - \alpha)\eta | \mathcal{F}_t] - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]) \geq 0$$

$$P(1_A E_g[\alpha \xi + (1 - \alpha)\eta | \mathcal{F}_t] - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]) > 0) > 0$$

$$g(t, 0) \equiv 0, A \in \mathcal{F}_t \Rightarrow E_g[1_A X | \mathcal{F}_t] = 1_A E_g[X | \mathcal{F}_t], \forall X \in L^2(\mathcal{F}_t) \quad (3.13)$$

g i.d. of y , $A \in \mathcal{F}_t + \text{Def 1.2} + \text{Lem 1.1 (iv) (iii)} + (3.13)$

$$\Rightarrow E_g[1_A (\alpha \xi + (1 - \alpha)\eta) - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t])] \\ = E_g[1_A (\alpha \xi + (1 - \alpha)\eta) - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]) | \mathcal{F}_t].$$

$$= E_g[1_A (\alpha \xi + (1 - \alpha)\eta) | \mathcal{F}_t] - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]).$$

$$= E_g[1_A E_g[\alpha \xi + (1 - \alpha)\eta | \mathcal{F}_t] - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t])] > 0.$$

$$E_g[\cdot] \text{ convex, } g \text{ is i.d. of } y. \quad (3.14)$$

$$+ \text{Def 1.2} + \text{Lem 1.2 (iv)} + (3.13) \Rightarrow$$

$$E_g[1_A (\alpha \xi + (1 - \alpha)\eta) - 1_A (\alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t])] \\ = E_g[\alpha (1_A \xi - 1_A E_g[\xi | \mathcal{F}_t]) + (1 - \alpha) (1_A \eta - 1_A E_g[\eta | \mathcal{F}_t])]$$

$$\leq \alpha E_g[1_A \xi - 1_A E_g[\xi | \mathcal{F}_t]] + (1 - \alpha) E_g[1_A \eta - 1_A E_g[\eta | \mathcal{F}_t]]$$

$$= \alpha E_g[1_A (\xi - E_g[\xi | \mathcal{F}_t]) | \mathcal{F}_t] + (1 - \alpha) E_g[1_A (\eta - E_g[\eta | \mathcal{F}_t]) | \mathcal{F}_t].$$

$$= \alpha E_g[1_A (\xi - E_g[\xi | \mathcal{F}_t]) - 1_A E_g[\xi | \mathcal{F}_t]] + (1 - \alpha) E_g[1_A (\eta - E_g[\eta | \mathcal{F}_t]) - 1_A E_g[\eta | \mathcal{F}_t]]$$

$$= \alpha E_g[1_A E_g[\xi | \mathcal{F}_t] - 1_A E_g[\xi | \mathcal{F}_t]] + (1 - \alpha) E_g[1_A E_g[\eta | \mathcal{F}_t] - 1_A E_g[\eta | \mathcal{F}_t]]$$

$$= 0 + 0 = 0. \quad (3.15)$$

$$(H3) \mathcal{E}_{g_1, r}[\xi] = \mathcal{E}_{g_2, r}[\xi], \forall r \in [0, T], \xi \in L^2(\Omega, \mathcal{F}_T, P).$$

Proposition 3.4. Let two generators g_1, g_2 satisfy assumptions (A1) and (A3) and hypothesis (H3). Let $0 \leq r \leq s \leq T$. Then for $\forall \xi \in L^2(\Omega, \mathcal{F}_s, P)$, P -a.s., we have:

$$\mathcal{E}_{g_1, s}[\xi | \mathcal{F}_r] = \mathcal{E}_{g_2, s}[\xi | \mathcal{F}_r].$$

$$\Rightarrow P\text{-a.s.}, \forall t \in [0, T], E_{g^c}[\xi | \mathcal{F}_t] = E_g[\xi | \mathcal{F}_t]. \quad (3.2)$$

$$\forall (y, z) \in \mathbb{R} \times \mathbb{R}^d, \forall t \in S_y^z(g^c) \cap S_y^z(g).$$

$$(3.2) \Rightarrow P\text{-a.s. } g^c(t, y, z) = g(t, y, z) \quad (3.3)$$

$$\text{Lem 2.1} \Rightarrow \lambda([0, T] \setminus (S_y^z(g^c) \cap S_y^z(g))) = 0 \quad (3.4)$$

$$\Rightarrow dP \times dt\text{-a.s.}, g^c(t, y, z) = g(t, y, z). \quad (3.5)$$

g and g^c are L w.r.t. $(y, z) \Rightarrow$

$$dP \times dt\text{-a.s.}, \forall y \in \mathbb{R}, z \in \mathbb{R}^d, g^c(t, y, z) = g(t, y, z) \quad (3.6)$$

That is, $\forall c \in \mathbb{R}, g^c = g \Rightarrow \forall y \in \mathbb{R}$.

$$(3.7) dP \times dt\text{-a.s. } \forall z \in \mathbb{R}^d, g(t, y, z) = g(t, 0, z) \quad \square.$$

• Convexity for g -expectation.

Thm 3.2 g - (A1) (A3). The following 3 statements are equivalent:

- $E_g[\cdot]$ is convex.
- $\forall t \in [0, T], E_g[\cdot | \mathcal{F}_t]$ is convex, that is,
 $\forall \xi, \eta \in L^2(\mathcal{F}_T), \alpha \in [0, 1]$,
 $E_g[\alpha \xi + (1 - \alpha)\eta | \mathcal{F}_t] \leq \alpha E_g[\xi | \mathcal{F}_t] + (1 - \alpha) E_g[\eta | \mathcal{F}_t]$, P -a.s.
- g i.d. of y and convex w.r.t. z , that is,
 $\forall z_1, z_2 \in \mathbb{R}^d, \alpha \in [0, 1]$, $dP \times dt$ -a.s.,
 $g(t, \alpha z_1 + (1 - \alpha)z_2) \leq \alpha g(t, z_1) + (1 - \alpha)g(t, z_2)$.

Proof. (iii) $\Rightarrow$ (ii): Prop 3.5 in KPA; (ii) $\Rightarrow$ (i): trivial.

(i) $\Rightarrow$ (iii): Step 1. Prove that convex $E_g[\cdot] \Rightarrow$ trans. inv.

(i) and Lem 1.1. $\Rightarrow \forall \xi \in L^2(\mathcal{F}_T), c \in \mathbb{R}, \alpha \in [0, 1]$,

$$E_g[\alpha \xi + (1 - \alpha)c] \leq \alpha E_g[\xi] + (1 - \alpha)c. \quad (3.8)$$

$$\Rightarrow E_g[(1 - \frac{1}{n})\xi + c] = E_g[(1 - \frac{1}{n})\xi + \frac{1}{n}(nc)] \leq E_g[(1 - \frac{1}{n})\xi] + c.$$

$$\forall \xi \in L^2(\mathcal{F}_T), c \in \mathbb{R}.$$

(3.15) is contradict to (3.14) with assumption $P(A) > 0$.

Therefore $P(A) = 0$. (3.12) holds.

Step 3: Prove that g is convex.

$$\forall z_1, z_2 \in \mathbb{R}^d, \alpha \in [0, 1], \text{ if } t \in S^{\alpha z_1 + (1 - \alpha)z_2}(g) \cap S^{z_1}(g) \cap S^{z_2}(g)$$

By (3.12). We deduce that

$$P\text{-a.s. } g(t, \alpha z_1 + (1 - \alpha)z_2) \leq \alpha g(t, z_1) + (1 - \alpha)g(t, z_2).$$

$$\forall z_1, z_2 \in \mathbb{R}^d, \alpha \in [0, 1]. \text{ By Representation Lemma.}$$

$$\lambda([0, T] \setminus (S^{\alpha z_1 + (1 - \alpha)z_2}(g) \cap S^{z_1}(g) \cap S^{z_2}(g))) = 0.$$

$$\Rightarrow \forall z_1, z_2 \in \mathbb{R}^d, \alpha \in [0, 1].$$

$$dP \times dt\text{-a.s.}$$

$$g(t, \alpha z_1 + (1 - \alpha)z_2) \leq \alpha g(t, z_1) + (1 - \alpha)g(t, z_2). \quad (3.16) \quad \square.$$

Similarly to Thm 3.2, We have

THEOREM 3.3 (Subadditivity for g -expectation). Let (A1) and (A3) hold for g . Then the following three statements are equivalent:

- $E_g[\cdot]$ is subadditive. (Subadditivity.)
- For any $t \in [0, T]$, $E_g[\cdot | \mathcal{F}_t]$ is subadditive, that is, $\forall \xi, \eta \in L^2(\mathcal{F}_T)$,

$$E_g[\xi + \eta | \mathcal{F}_t] \leq E_g[\xi | \mathcal{F}_t] + E_g[\eta | \mathcal{F}_t], \quad P\text{-a.s.}$$

- g is independent of y and g is subadditive with respect to z , that is, for any $z_1, z_2 \in \mathbb{R}^d$,

$$g(t, z_1 + z_2) \leq g(t, z_1) + g(t, z_2), \quad dP \times dt\text{-a.s.}$$

Thm 3.4 (Positive homogeneity for g -expectation)

g - (A1) (A3). The following 3 statements are equivalent:

- $E_g[\cdot]$ is positively homogeneous.
- $\forall t \in [0, T], E_g[\cdot | \mathcal{F}_t]$ is p.h.
- g is p.h. w.r.t. (y, z) .

Proof. (iii) $\Rightarrow$ (ii) : Prop 8. of Rosazza. (ii) $\Rightarrow$ (i) : trivial.

(i) $\Rightarrow$ (iii) : $\forall \alpha > 0$. Def: $\tilde{g}^\alpha(t, y, z) = \alpha g(t, \frac{y}{\alpha}, \frac{z}{\alpha})$.

$\tilde{g}^\alpha - (A1)(A2)(A3)$. $\forall \xi \in L^2(\mathcal{F}_T)$. We deduce that.

$$(Y, (\tilde{g}^\alpha, T, \alpha \xi), Z, (\tilde{g}^\alpha, T, \alpha \xi)) = \alpha (Y, (g, T, \xi), Z, (g, T, \xi))$$

$$\forall \alpha > 0, E_{\tilde{g}^\alpha}[\alpha \xi] = \alpha E_g[\xi], \forall \xi \in L^2(\mathcal{F}_T).$$

Combining with (i). $E_{\tilde{g}^\alpha}[\xi] = E_g[\xi], \forall \xi \in L^2(\mathcal{F}_T)$ (3.17)

Same as the argument as in (3.1) - (3.6), we have $\tilde{g}^\alpha = g$.

$$\Rightarrow (3.18) \quad dP \times dt - a.s. \quad \forall (y, z) \in \mathbb{R}^{1+d}, g(t, y, z) = \alpha g(t, \frac{y}{\alpha}, \frac{z}{\alpha})$$

Hence (iii) follows from (3.18) and (A1). $\square$

REMARK 3.2. Under (A1), (A3) and (A4), [21] proved that (ii) is equivalent to (iii) in Theorems 3.2-3.4; see Propositions 8-11 in [21].

Risk Measure via g -expectation.

Def 4.1 $\mathcal{G}$ - set of risks (IR-valued func. on Ω)

$\rho: \mathcal{G} \rightarrow \mathbb{R}$ - monetary risk measure of.

$\forall X, Y \in \mathcal{G}$, ρ satisfies

$$(1) \text{ Monotonicity : } X \leq Y \Rightarrow \rho(X) \geq \rho(Y)$$

$$(2) \text{ Translation invariance : } \rho(X+c) = \rho(X) - c, \forall c \in \mathbb{R}.$$

Def 4.2 A monetary risk measure is called a convex measure of risk if it satisfies:

$$(3) \text{ Convexity : } \rho(\lambda X + (1-\lambda)Y) \leq \lambda \rho(X) + (1-\lambda)\rho(Y), \forall \lambda \in [0,1]$$

$$(4) \text{ Positive homogeneity : } \rho(\lambda X) = \lambda \rho(X).$$

Def 4.3 $\mathcal{G} \subseteq L^0(\Omega, \mathcal{F}_T, P)$. A map system $(\rho_t)_{t \in [0, T]}$ is called a dynamic risk if $\forall X, Y \in \mathcal{G}, t \in [0, T]$:

$$1^\circ \rho_t: \mathcal{G} \rightarrow L^0(\Omega, \mathcal{F}_t, P)$$

$$2^\circ \rho_0 \text{ is a static monetary risk measure}$$

$$3^\circ \rho_T(X) = -X.$$

$$4^\circ \text{ Dynamic monotonicity : } X \leq Y \Rightarrow \rho_t(X) \geq \rho_t(Y)$$

$$5^\circ \text{ Dynamic translation invariance : } \rho_t(X+c) = \rho_t(X) + c, \forall c \in \mathbb{R}.$$

Def 4.4 A dynamic r.m. $(\rho_t)_{t \in [0, T]}$ is called a dynamic convex measure of risk if

$$6^\circ \text{ Dynamic convexity : } \forall X, Y \in \mathcal{G}, \lambda \in [0,1], t \in [0, T], P\text{-a.s.}$$

$$\rho_t(\lambda X + (1-\lambda)Y) \leq \lambda \rho_t(X) + (1-\lambda)\rho_t(Y).$$

$\sim$ is called dynamic coherent risk measure if

$$7^\circ \forall X \in \mathcal{G}, \lambda \geq 0, t \in [0, T] \quad P\text{-a.s.} \quad \rho_t(\lambda X) = \lambda \rho_t(X)$$

Thm 4.1 $g - (A1)(A3) \quad \mathcal{G} = L^2(\mathcal{F}_T)$.

ρ^g and $(\rho_t^g)_{t \in [0, T]}$ are defined as below

$$\rho^g(\xi) := E_g[-\xi], \quad \rho_t^g(\xi) := E_g[-\xi | \mathcal{F}_t] \quad \forall t \in [0, T].$$

The following statements are equivalent:

$$(i) \rho^g \text{ is a convex measure of risk.}$$

$$(ii) (\rho_t^g)_{t \in [0, T]} \text{ is a dynamic convex measure of risk.}$$

$$(iii) E_g \text{ is convex.}$$

$$(iv) g \text{ i.d. of } y \text{ and convex w.r.t. } z.$$

Thm 4.2 $g, \mathcal{G}, \rho^g, (\rho_t^g)_{t \in [0, T]}$ remain as in Thm 4.1.

The following statements are equivalent:

$$(i) \rho^g \text{ is a convex measure of risk.}$$

$$(ii) (\rho_t^g)_{t \in [0, T]} \text{ is a dynamic convex measure of risk}$$

$$(iii) E_g \text{ is sublinear (sub-additive and p.h.)}$$

$$(iv) g \text{ i.d. of } y \text{ and sublinear w.r.t. } z.$$

By Theorems 3.1-3.4 and Definitions 4.1-4.4, we can obtain the following Theorems 4.1 and 4.2 immediately.

REMARK 4.1. Rosazza Gianin [21] proved that (ii) is equivalent to (iv) in Theorems 4.1 and 4.2 under assumptions (A1), (A3) and (A4).